

BIRLA INSTITUTE OF TECHNOLOGY, MESRA, RANCHI
(END SEMESTER EXAMINATION MO-2023)

CLASS: B.TECH.
BRANCH: CSE/AIML/ECE/EEE

SEMESTER: I
SESSION: MO/2023

SUBJECT: EC101 BASICS OF ELECTRONICS AND COMMUNICATION ENGINEERING

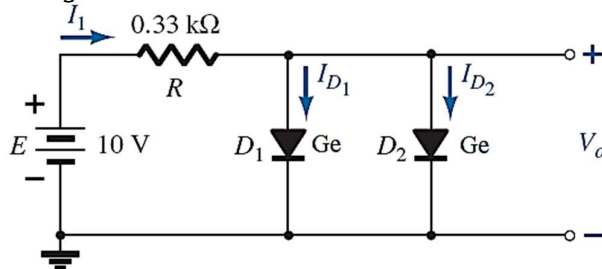
TIME: 3 HOURS

FULL MARKS: 50

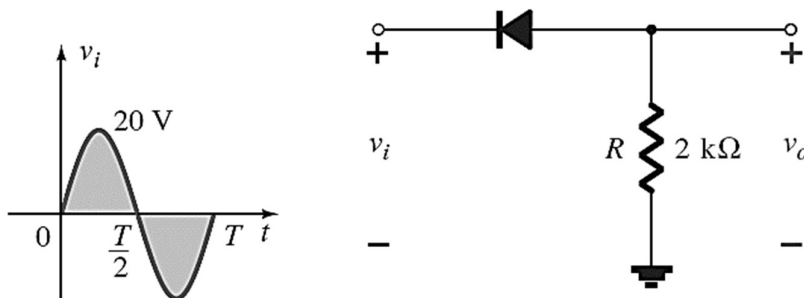
INSTRUCTIONS:

1. The question paper contains 5 questions each of 10 marks and total 50 marks.
2. Attempt all questions.
3. The missing data, if any, may be assumed suitably.
4. Tables/Data handbook/Graph paper etc., if applicable, will be supplied to the candidates.

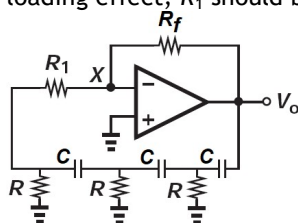
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|--|-----|---------|---------|
| Q1 (a) Determine (i) V_o , (ii) I_1 , (iii) I_{D1} , and (iv) I_{D2} for the parallel diode configuration given in the figure. | [5] | CO
1 | BL
3 |
|--|-----|---------|---------|



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|--|-----|---|---|
| Q1 (b) Sketch the output v_o and determine the average out voltage (V_{avg}), i.e., DC out voltage (V_{dc}) for the half-wave rectifier with a silicon diode having cutin voltage $\gamma = 0.7$ V and v_i [i.e., the output voltage of secondary winding of the transformer (not shown)] as shown in the given figure. Also, write down its Peak Inverse Voltage (PIV). | [5] | 1 | 3 |
|--|-----|---|---|



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|--|-----|---|------|
| Q2 (a) For a PNP BJT the base current and collector current are respectively 45 μ A and 5.45 mA. Determine (i) values of transistor α , β and I_E ; (ii) Base current required to make collector current of 10 mA. | [5] | 2 | 3 |
| Q2 (b) (i) Sketch the drain characteristics (I_D - V_{DS}) of a p-channel enhancement MOSFET for $V_{GS} = -4$ V, -12 V, -20 V. (ii) Sketch the transfer curve (i.e., I_D - V_{GS}) (for $V_{DS} = 10$ V) of a p-channel enhancement-type MOSFET. | [5] | 2 | 3 |
| Q3 (a) Design an OP AMP-based 3-stage RC phase-shift oscillator as given in the figure. Assume that the frequency of oscillation of the oscillator is 10 kHz and the value of C is 0.1 μ F. To complete the design, estimate the values of R_1 , R_f , and R. Assume that, to avoid the loading effect, R_1 should be much greater than R (i.e., $R_1 \gg R$, at least 10 times). | [5] | 3 | 5, 6 |



Q3	(b)	Schematize the op-amp-based adder circuit, or summing amplifier circuit, and deduce the expression of its output voltage (v_o).	[5]	3	6, 4
Q4	(a)	Design a 2-input XOR gate with a minimum number of 2-input NOR gates. Design a half adder with a minimum number of 2-input NOR gates.	[5]	4	6
Q4	(b)	Schematize a Logic Diagram of a 4-bit Ripple Carry Adder/subtractor and briefly explain it.	[5]	4	6
Q5	(a)	What are three distinct modes of transmission for any communications link? Briefly explain each with at least one example.	[5]	5	1, 2
Q5	(b)	A carrier signal $v_c(t) = 5 \sin [(2\pi \times 10^6)t]$ is amplitude modulated by a modulating sinusoidal signal $v_m(t) = \sin [(4\pi \times 10^3)t]$. Determine the amplitude and frequency of the carrier signal and the modulating signal. Also, write the expression for the resulting AM signal.	[5]	5	1, 3

.....15/12/2023 M:.....